

# **Forecasting Short-Term Gold ETF Returns and Volatility: ARIMA-GARCH Modeling and Monte Carlo Simulation**

**MFIN8852.01**

**Financial Econometrics**

**Sofia Kolaro, Utkarsh Maini, Elizabeth Murray & Kiki Stoker**

*Group project. The paper was written in full by Sofia Kolaro.*



## Introduction

Forecasting financial asset returns and the volatility that drives their risk is a core element of modern investment analysis and risk management. Gold is widely regarded as an international safe-haven asset, providing an ideal case for assessing the effectiveness of time-series models designed to capture both return and volatility dynamics. This project analyzes the SPDR Gold Trust (GLD) using daily data from January 2010 through November 2025, applying an ARIMA–GARCH framework combined with a Monte Carlo simulation. The primary goal is to model the conditional mean and variance of gold returns and evaluate the forecasting performance of these models through out-of-sample prediction intervals.

## Methods

### Exploratory Data Analysis

The dataset, sourced from Yahoo Finance, consists of daily GLD closing prices from January 1, 2010, to November 14, 2025. For model estimation, the training sample includes observations up to, but excluding, five trading days (November 10–14, 2025). These days are intentionally excluded from model fitting and reserved as an out-of-sample testing window to evaluate the model’s forecasting accuracy. Once the ARIMA–GARCH model is estimated, it is used to generate forecasts for the testing period, which are then compared with realized returns. The same model is later extended to the entire dataset to produce forecasts for the subsequent five-trading-day period (November 17–21, 2025).

A five-day testing horizon is appropriate, given the short-run predictive nature of daily return and volatility models. In particular, GARCH models are known to produce their most reliable forecasts over near-term windows, and one-week intervals are commonly used in practice for short-horizon risk management.

Before estimation, raw prices were transformed into logarithmic returns to stabilize variance and ensure additivity over time, which are properties crucial for modeling data characterized by heteroscedasticity and long-run trends. The resulting log-return series provides stationarity, making it suitable for ARIMA and GARCH models.

### Descriptive Analysis of GLD Price Levels

Figure 1 plots the evolution of GLD prices over the full sample period. The series displays significant non-stationarity, moving through extended low-volatility regimes followed by strong upward movements (most notably during 2020 and again in 2024–2025). This

persistent trend structure confirms that modeling prices directly would be inappropriate for stationarity-based time-series models.

Figure 1



Figure 2

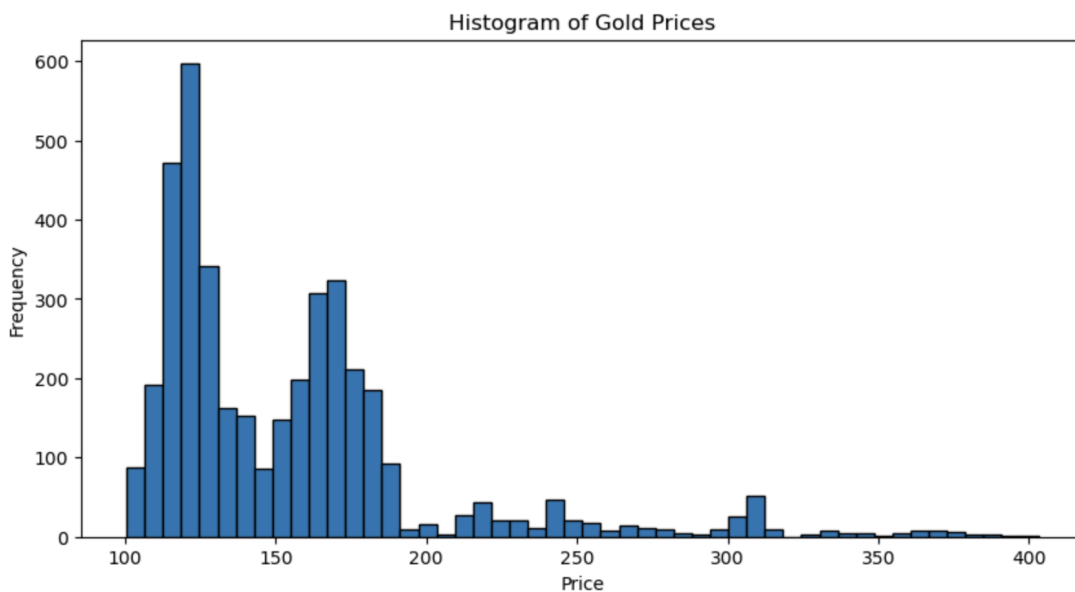
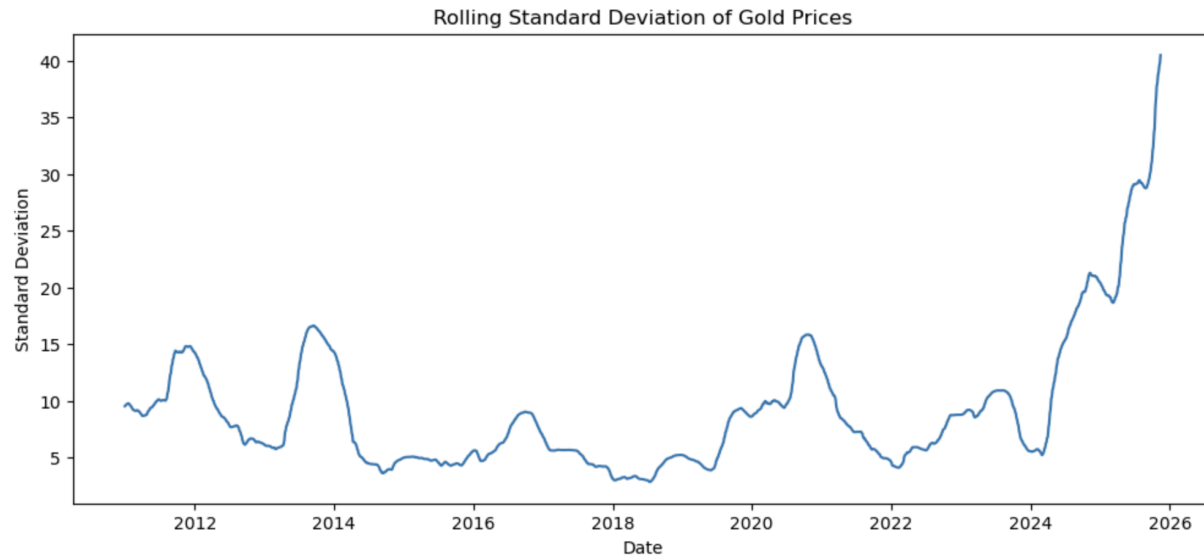


Figure 2, displays the histogram of GLD prices, showing a clearly non-Gaussian distribution with clustering at lower price levels, occasional extreme spikes, and pronounced fat tails. The right tail extends substantially farther than the left, indicating positive skewness, a feature commonly observed in commodity markets where abrupt upward shifts occur during

periods of heightened uncertainty. This asymmetry and tail heaviness reinforce that raw prices violate normality and stationarity.

**Figure 3**



**Figure 3** extends this idea by showing the rolling standard deviation of GLD prices. Volatility appears to be far from constant and exhibits strong clustering, with noticeable spikes around major market events. For example, the surge around 2020 corresponds to the outbreak of the COVID-19 pandemic, while the more recent increase from 2024 reflects increased macroeconomic uncertainty and geopolitical tensions. These episodes demonstrate that gold volatility tends to rise during periods of global stress, reinforcing the use of GARCH models to capture time-varying risk.

### **Ensuring Stationarity: Log Returns and ADF Test**

The patterns observed in the price data, from volatility clustering to non-normality, indicate that GLD prices need to be transformed prior to ARIMA and GARCH modeling. To address these issues, the price series was converted into logarithmic returns, providing more stable statistical properties for capturing short-run dynamics in time series. After constructing the log-return dataset, the Augmented Dickey-Fuller test was applied to formally test for stationarity. With a p-value of 0.00, the test strongly rejects the null hypothesis of a unit root, confirming that GLD log returns are stationary and ready for ARIMA–GARCH modeling.

### **Optimal ARIMA (p, d, q) Order Identification**

To identify the most appropriate ARIMA specification, an iterative loop was implemented in Python to find p, d, and q parameters with the lowest Akaike Information

Criterion score. The differencing parameter was fixed at  $d = 0$ , as the log-return series was already stationary according to the Augmented Dickey–Fuller test, making additional differencing unnecessary.

The loop returned ARIMA(0, 0, 0) as the model with the best fit, achieving the lowest AIC value of 11288.51. This result is consistent with typical asset return behavior, where daily returns generally exhibit little to no autocorrelation and often follow a white-noise process. Due to this lack of autocorrelation, including autoregressive (AR) or moving average (MA) terms does not significantly improve the model’s fit. The full model summary is presented in the figure below.

**Figure 4**

| SARIMAX Results         |                  |                   |                   |       |        |        |
|-------------------------|------------------|-------------------|-------------------|-------|--------|--------|
| =====                   |                  |                   |                   |       |        |        |
| Dep. Variable:          | GLD              | No. Observations: | 3987              |       |        |        |
| Model:                  | ARIMA            | Log Likelihood    | -5642.257         |       |        |        |
| Date:                   | Sat, 15 Nov 2025 | AIC               | 11288.515         |       |        |        |
| Time:                   | 19:57:59         | BIC               | 11301.096         |       |        |        |
| Sample:                 | 0                | HQIC              | 11292.975         |       |        |        |
|                         | - 3987           |                   |                   |       |        |        |
| Covariance Type:        | opg              |                   |                   |       |        |        |
| =====                   |                  |                   |                   |       |        |        |
|                         | coef             | std err           | z                 | P> z  | [0.025 | 0.975] |
| -----                   |                  |                   |                   |       |        |        |
| const                   | 0.0304           | 0.016             | 1.887             | 0.059 | -0.001 | 0.062  |
| sigma2                  | 0.9925           | 0.012             | 80.525            | 0.000 | 0.968  | 1.017  |
| =====                   |                  |                   |                   |       |        |        |
| =                       |                  |                   |                   |       |        |        |
| Ljung-Box (L1) (Q):     |                  | 1.00              | Jarque-Bera (JB): |       | 3935.8 |        |
| 6                       |                  |                   |                   |       |        |        |
| Prob(Q):                |                  | 0.32              | Prob(JB):         |       | 0.0    |        |
| 0                       |                  |                   |                   |       |        |        |
| Heteroskedasticity (H): |                  | 0.78              | Skew:             |       | -0.5   |        |
| 1                       |                  |                   |                   |       |        |        |
| Prob(H) (two-sided):    |                  | 0.00              | Kurtosis:         |       | 7.7    |        |
| 6                       |                  |                   |                   |       |        |        |
| =====                   |                  |                   |                   |       |        |        |

## ARIMA (0, 0, 0) Model Interpretation

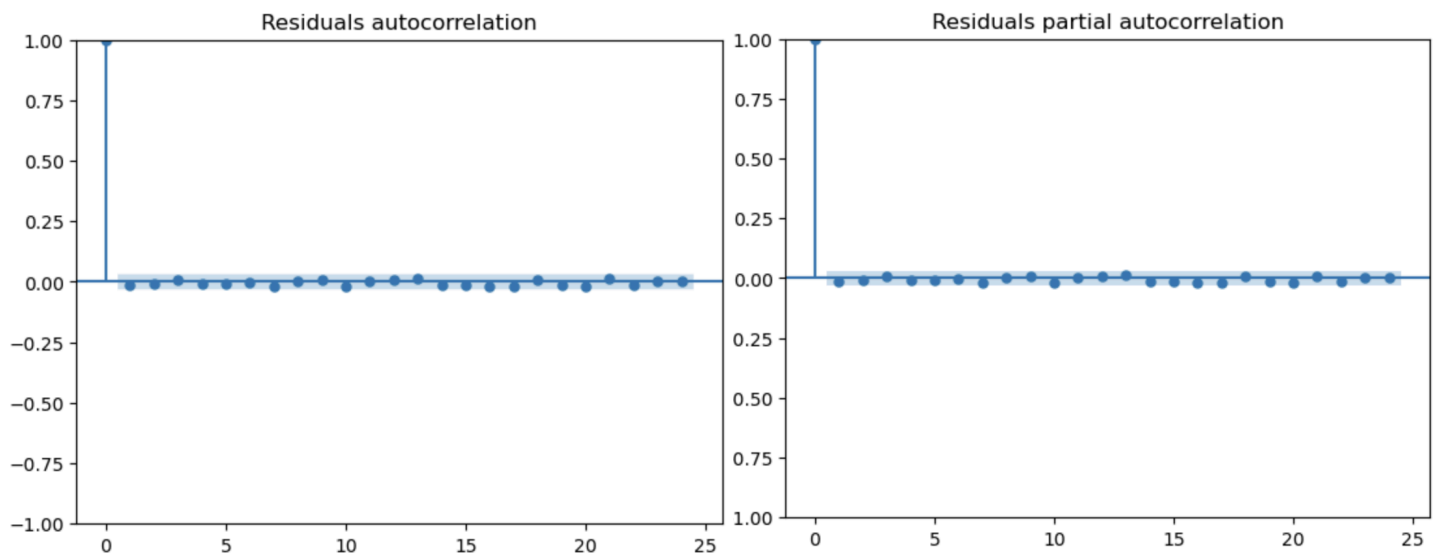
Figure 4 shows that the ARIMA(0,0,0) model confirms that daily GLD log returns behave much like a white-noise process, consistent with financial theory. The estimated constant is small and only marginally significant. The Ljung-Box statistic ( $p = 0.32$ ) detects no meaningful autocorrelation in the residuals, indicating a lack of predictable structure in the mean. The Jarque–Bera test strongly rejects normality, and the high kurtosis value (7.7) points to heavy tails, suggesting that extreme return realizations occur more frequently than under a normal distribution. Despite the lack of serial correlation in the mean, the significant heteroskedasticity test ( $p = 0.00$ ) indicates volatility clustering in the residuals, providing a clear justification for further developing the mean equation with a GARCH (1,1) model to capture time-dependent conditional variance.

The mean equation,  $r_t = 0.0304 + \varepsilon_t$ , where  $\varepsilon_t \sim \text{WN}(0, 0.9925)$ , implies an expected daily return of approximately 0.0308%. Over a full year (252 trading days), this daily return compounds to an expected annual gain of about 7.66%. However, the p-value of 0.059 shows the mean is only borderline significant, meaning the true expected daily return may be close to 0.

### Autocorrelation Structure: ACF and PACF

To further assess whether the ARIMA (0, 0, 0) model adequately captures the mean dynamics of GLD log returns, additional residual diagnostics were conducted. In particular, the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots provide a visual check for any remaining serial dependence in the residuals. If the ARIMA specification is appropriate, both plots should show no significant spikes beyond the confidence bands.

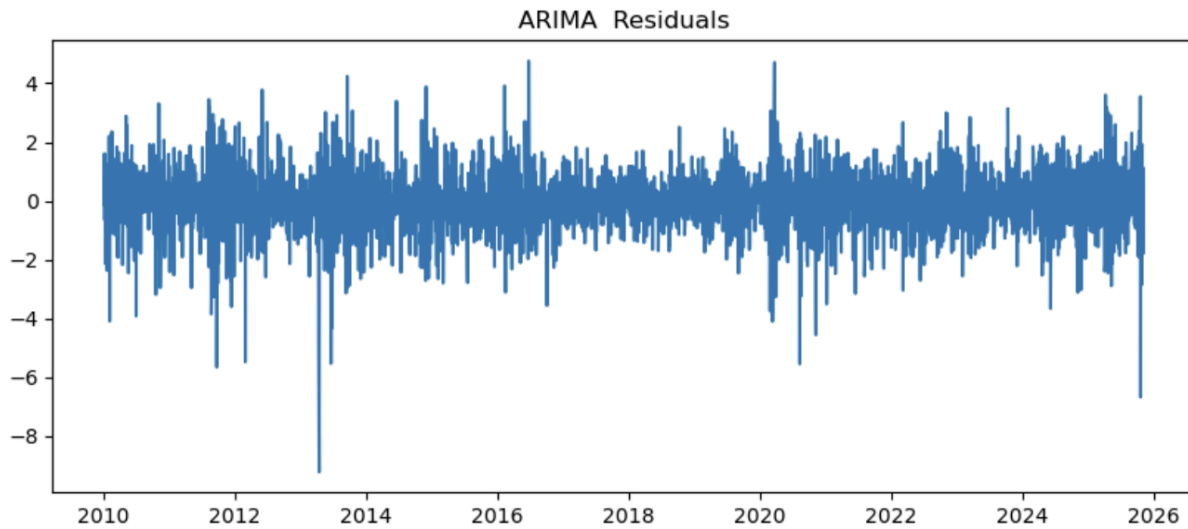
Figures 5 and 6



Consistent with the Ljung-Box results reported earlier, the autocorrelation structure is flat across all lags. This confirms that the mean process contains no remaining predictable structure and that an ARIMA(0,0,0) specification is sufficient for modeling the conditional mean.

Figure 7 shows the ARIMA residuals over time, revealing patterns of calm periods followed by heightened volatility. These alternating regimes of low and high standard deviation are characteristic of volatility clustering, a feature of financial returns that cannot be captured by the ARIMA model alone and one that requires the use of a GARCH model.

Figure 7



### Testing for Volatility Clustering with Engle's ARCH LM Test

The GARCH (1, 1) model is widely regarded as the standard for modeling conditional variance in financial time series due to its simplicity and effectiveness. In this framework, current volatility depends on both the most recent shock (the ARCH term) and the previous period's volatility (the GARCH term). In our results, the ARCH LM statistic is 145.58 with a p-value of 0.00, offering strong evidence against homoscedasticity and supporting the suitability of the GARCH (1, 1) specification.

### Modeling Volatility in Gold Returns Using GARCH(1,1)

Figure 8

| Zero Mean - GARCH Model Results |                    |                   |          |            |                        |
|---------------------------------|--------------------|-------------------|----------|------------|------------------------|
| =====                           |                    |                   |          |            |                        |
| Dep. Variable:                  | None               | R-squared:        | 0.000    |            |                        |
| Mean Model:                     | Zero Mean          | Adj. R-squared:   | 0.000    |            |                        |
| Vol Model:                      | GARCH              | Log-Likelihood:   | -5411.20 |            |                        |
| Distribution:                   | Normal             | AIC:              | 10828.4  |            |                        |
| Method:                         | Maximum Likelihood | BIC:              | 10847.3  |            |                        |
|                                 |                    | No. Observations: | 3987     |            |                        |
| Date:                           | Sat, Nov 15 2025   | Df Residuals:     | 3987     |            |                        |
| Time:                           | 20:44:59           | Df Model:         | 0        |            |                        |
| Volatility Model                |                    |                   |          |            |                        |
| =====                           |                    |                   |          |            |                        |
|                                 | coef               | std err           | t        | P> t       | 95.0% Conf. Int.       |
| omega                           | 0.0239             | 1.165e-02         | 2.054    | 3.997e-02  | [1.096e-03, 4.675e-02] |
| alpha[1]                        | 0.0648             | 1.887e-02         | 3.437    | 5.884e-04  | [2.787e-02, 0.102]     |
| beta[1]                         | 0.9120             | 2.784e-02         | 32.756   | 2.455e-235 | [ 0.857, 0.967]        |

Figure 8 presents the estimation results from the GARCH (1, 1) model, given by:

$$\sigma_t^2 = 0.0239 + 0.0648\varepsilon_{t-1}^2 + 0.9120\sigma_{t-1}^2.$$

The constant term,  $\omega = 0.0239$ , sets a lower bound on the conditional variance, since the ARCH and GARCH terms are non-negative. This confirms that gold returns exhibit a non-zero volatility floor, while volatility itself reverts to the long-run variance estimated below.

The ARCH coefficient,  $\alpha_1 = 0.0648$ , is significant and captures the immediate impact of the previous period's unexpected market shock. This indicates that about 6.5% of yesterday's shock carries over into today's volatility. The GARCH coefficient,  $\beta_1 = 0.9120$ , is also highly significant and close to 1, implying strong volatility persistence. This means that periods of high or low volatility tend to continue over multiple days. Moreover, since the sum of these two terms is close to 1 (0.9768), volatility shocks are very persistent, approaching an Integrated GARCH (IGARCH) process, so volatility may take a very long time to return to its long-run level.

### **Estimating Long-Run Unconditional Volatility**

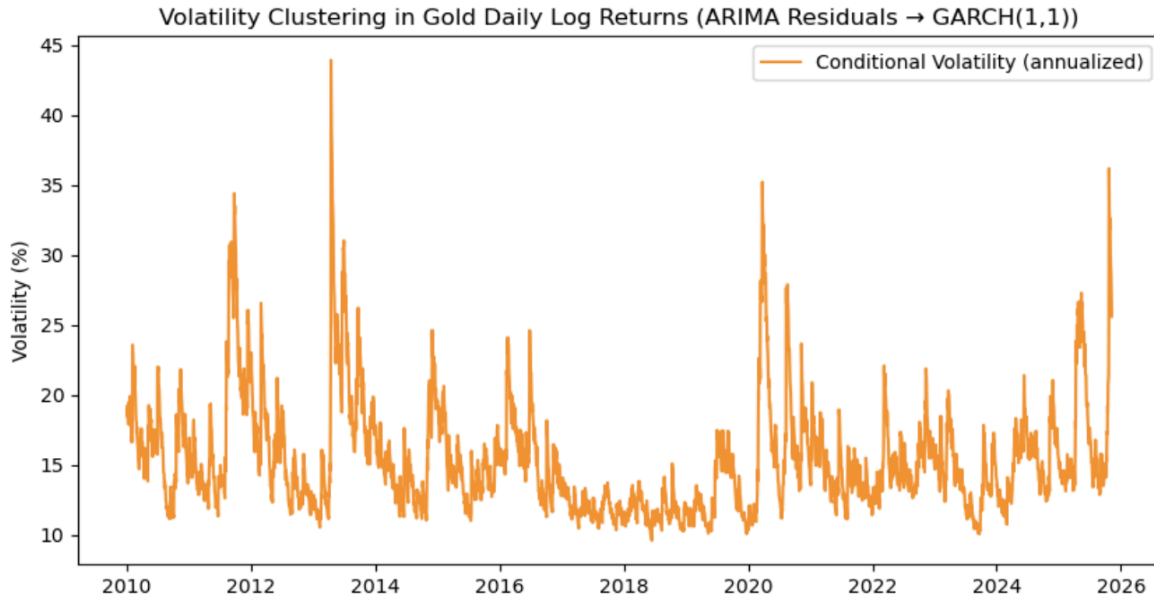
Given the estimated parameters, it is also useful to compute the long-run unconditional variance implied by the GARCH (1, 1) model to estimate the average volatility level toward which the process eventually converges. While the conditional volatility fluctuates daily, the unconditional variance determines the model's long-run equilibrium.

The long-run unconditional variance is approximately 1.0302, which indicates that GLD log returns maintain a relatively high baseline level of volatility. Taking the square root of this value yields an unconditional standard deviation of about 1.015%, meaning that, on average, daily GLD log returns are expected to fluctuate by roughly 1.015% around their mean in the long run (roughly 16.11% annually). This provides a benchmark for the magnitude of typical daily movements and confirms that the GARCH model captures the elevated and persistent volatility characteristic of financial time series.

### **Estimating Conditional Volatility**

With the GARCH (1, 1) parameters and long-run variance established, we then estimate the full sequence of conditional volatilities over the entire sample period. Using the fitted GARCH model, a conditional variance estimate is generated for each day based on past shocks and past volatility, producing a time series of daily volatility that reflects how risk evolves through time. To make these volatility estimates more interpretable, the daily conditional variances were annualized using 252 trading days. This produces a smoother and more intuitive measure of GLD's evolving risk profile. The resulting annualized conditional volatilities are plotted below to illustrate the pronounced volatility clustering captured by the GARCH model.

**Figure 9**



Throughout the sample period, volatility fluctuates between approximately 10% and 40%, demonstrating that gold returns can shift between relatively stable periods and intense turbulence. The slow and smooth decay following each spike indicates that volatility shocks dissipate gradually over time. Overall, the volatility plot confirms the GARCH model's effectiveness in capturing gold's time-varying volatility and the prolonged calm periods followed by bursts of market instability, a hallmark of financial assets.

### **Dynamic Risk Bands Based on ARIMA Mean and GARCH Volatility Forecasts**

By combining the ARIMA model's forecasts for the conditional mean with the GARCH model's forecasts for conditional volatility, we construct dynamic prediction intervals for the out-of-sample five-day trading period. The ARIMA (0, 0, 0) model provides the forecasted mean return  $\mu_t$ , while the GARCH (1, 1) model generates the corresponding forecast of the time-varying conditional volatility  $\sigma_t$ . Using these forecasts, 95% volatility-adjusted prediction intervals are computed as:

$$\text{Lower}_t = \mu_t - 1.96\sigma_t \text{ and } \text{Upper}_t = \mu_t + 1.96\sigma_t$$

These intervals naturally widen during periods of elevated volatility and tighten during calmer periods, mirroring the time-varying risk captured by the GARCH model. This approach yields forecast bands that adapt to changing market conditions, offering a more realistic representation of uncertainty.

Figure 10

|            | Lower     | Mean    | Upper    | Actual    |
|------------|-----------|---------|----------|-----------|
| Date       |           |         |          |           |
| 2025-11-10 | -3.017495 | 0.03035 | 3.078196 | 2.697403  |
| 2025-11-11 | -2.997210 | 0.03035 | 3.057911 | 0.393008  |
| 2025-11-12 | -2.977263 | 0.03035 | 3.037964 | 1.598236  |
| 2025-11-13 | -2.957649 | 0.03035 | 3.018350 | -0.811594 |
| 2025-11-14 | -2.938364 | 0.03035 | 2.999065 | -1.821276 |

Hit ratio: 1.0  
Days outside interval: 0

Figure 10 indicates that the model achieves a hit ratio of 1.0, with all five realized returns falling inside the 95% prediction intervals. This full coverage indicates that the ARIMA–GARCH forecast bands were sufficiently wide to accommodate the short-run volatility observed during the test window and is consistent with well-calibrated intervals that captured the underlying market uncertainty. However, with only five out-of-sample observations, a correctly specified 95% interval would contain all five realized returns roughly 77% of the time ( $0.95^5$ ), so a longer testing window would be needed to evaluate calibration more rigorously.

Figure 11

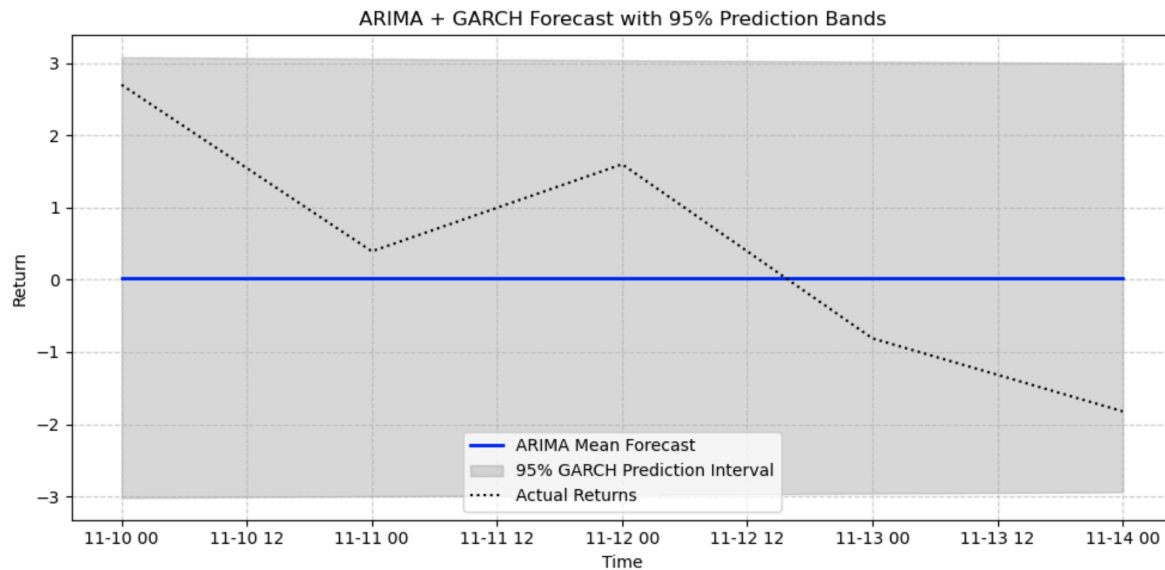


Figure 11 shows the out-of-sample period covering November 10 to November 14, 2025. The ARIMA (0, 0, 0) model produced a constant mean forecast, reflecting the absence of predictable trends in daily gold returns. Meanwhile, the GARCH (1, 1) model generated time-varying volatility forecasts that adjusted to changing market conditions. The resulting 95% prediction bands successfully encompassed all realized returns within this forecast window.

Dynamic Risk Bands Using a Monte Carlo Simulation

As an alternative to the analytical ARIMA–GARCH prediction intervals, a Monte Carlo simulation was implemented to generate an additional set of dynamic risk bands for the out-of-sample testing window. From this distribution, the 2.5th percentile, 50th percentile (median), and 97.5th percentile were extracted to form the lower, central, and upper bounds of the simulation-based prediction intervals.

Figure 12 shows that for all five days in the out-of-sample period, the realized GLD returns fall inside the Monte Carlo 95% interval. This results in a hit ratio of 1.0, consistent with the simulated risk bands capturing the short-run uncertainty in gold returns. Unlike the analytical ARIMA–GARCH intervals, the median return forecast here represents the typical outcome across all simulated paths, rather than an ARIMA mean forecast. The close alignment between actual returns and the Monte-Carlo-derived intervals suggests that the simulation approach provides a robust, distribution-driven measure of forecast uncertainty.

Figure 12

Hit ratio: 1.0  
Days outside interval: 0

|            | Lower     | Median   | Upper    | GLD       |
|------------|-----------|----------|----------|-----------|
| Date       |           |          |          |           |
| 2025-11-10 | -2.983021 | 0.028599 | 2.959604 | 2.697403  |
| 2025-11-11 | -2.995576 | 0.019692 | 2.984976 | 0.393008  |
| 2025-11-12 | -2.944794 | 0.021260 | 2.985217 | 1.598236  |
| 2025-11-13 | -3.012184 | 0.020083 | 3.056944 | -0.811594 |
| 2025-11-14 | -2.959898 | 0.007196 | 2.992906 | -1.821276 |

Figure 13

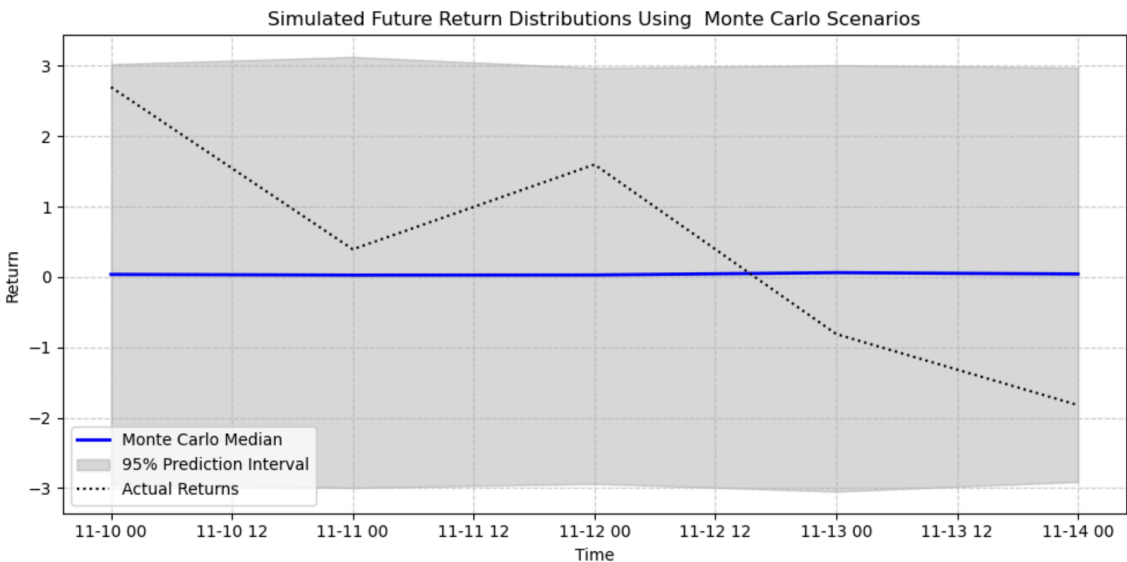


Figure 13 offers a visual representation of Monte Carlo intervals. The shaded region represents the 95% prediction interval, capturing the range in which most simulated outcomes fall. The blue line plots the median simulated return, which remains close to zero, consistent with the ARIMA (0, 0, 0) mean forecast. The dotted black line shows the actual realized returns, all of which lie comfortably within the simulated bands.

### Forecasting Future GLD Prices

After validating the ARIMA–GARCH framework on the five-day out-of-sample window, the model was re-estimated on the full dataset and used to forecast gold returns for the subsequent five trading days (November 17–21, 2025). Once again, the ARIMA mean and the GARCH conditional volatility forecasts are combined to form dynamic 95% prediction intervals for log returns, which are then converted into implied price bands.

Figure 14

|                   | Lower     | Mean     | Upper    |                   | Lower Price | Mean Price | Upper Price |
|-------------------|-----------|----------|----------|-------------------|-------------|------------|-------------|
| <b>2025-11-17</b> | -2.974991 | 0.030827 | 3.036645 | <b>2025-11-17</b> | 364.939951  | 376.075907 | 387.551671  |
| <b>2025-11-18</b> | -2.955583 | 0.030827 | 3.017237 | <b>2025-11-18</b> | 354.311687  | 376.191859 | 399.423219  |
| <b>2025-11-19</b> | -2.936498 | 0.030827 | 2.998152 | <b>2025-11-19</b> | 344.058610  | 376.307846 | 411.579861  |
| <b>2025-11-20</b> | -2.917732 | 0.030827 | 2.979386 | <b>2025-11-20</b> | 334.164939  | 376.423869 | 424.026918  |
| <b>2025-11-21</b> | -2.899281 | 0.030827 | 2.960936 | <b>2025-11-21</b> | 324.615656  | 376.539928 | 436.769807  |

Figure 14 shows the forward-looking forecasts for November 17–21, including both return prediction intervals and their implied price ranges. The mean return forecast remains near zero across the horizon, consistent with the ARIMA (0, 0, 0) result that daily expected returns are largely unpredictable. The associated price bands widen over time: lower bounds drift downward while upper bounds rise. This divergence reflects the compounding effect of forecasted volatility and highlights that uncertainty grows with the prediction horizon even when the expected return is stable. Rather than producing a single point estimate, the model therefore delivers a realistic range of plausible future prices consistent with the conditional volatility structure captured by GARCH.

To assess short-horizon accuracy, the forecasted price bands were compared with realized GLD closing prices. The actual prices for November 17–21 (\$371.65, \$374.35, \$374.96, \$374.85, and \$374.27) all fall within the lower and upper price bounds shown in Figure 14, consistent with well-calibrated forecasts over the realized portion of the horizon. Overall, the forecasting results suggest that the ARIMA–GARCH model produces return and price intervals that meaningfully reflect near-term market risk and uncertainty.

## Conclusion

Combining ARIMA and GARCH models, supported by Monte Carlo simulation, provides a robust and flexible framework for forecasting gold returns and quantifying short-run risk. The ARIMA (0, 0, 0) model confirms the inherent unpredictability of daily gold returns, while the GARCH (1, 1) specification successfully captures the persistent volatility clustering and time-varying risk that exemplify financial markets.

The predicted risk intervals, derived analytically and through simulation, achieved full coverage in the five-day out-of-sample test period. While a longer evaluation window would be needed to confirm their calibration, this result is consistent with the model's ability to adjust forecasting uncertainty in response to shifts in volatility, expanding during turbulent periods and contracting during calmer ones. Converting return forecasts into price intervals for the forward-looking period offers a practical interpretation of the results, allowing for direct comparison with observed GLD price movements.

This approach offers valuable insights for investors and risk managers seeking to anticipate short-term fluctuations in gold prices and manage risk under dynamic market conditions. While price direction remains unpredictable for longer horizons, the model indicates that gold is in a period of elevated volatility, implying that future price movements are likely to be larger and more volatile than usual. This information allows portfolio managers and investors to adjust hedging strategies, liquidity buffers, and exposure levels in anticipation of heightened uncertainty.